

SQL Queries & CTEs of the Project

1) Finding the average amount paid by the top 5 customers

1.1) Outer statement/main query for the average amount paid

```
SELECT
  B.customer_id,
  B.first_name AS customer_first_name,
  B.last_name AS customer_last_name,
  D.city,
  E.country,
  AVG(A.amount) AS average_amount_paid
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE D.city IN (
  SELECT city
  FROM customer A
  INNER JOIN address B ON A.address_id = B.address_id
  INNER JOIN city C ON B.city_id = C.city_id
  INNER JOIN country D ON C.country_ID = D.country_ID
  WHERE D.country IN (
    SELECT country
    FROM customer A
    INNER JOIN address B ON A.address_id = B.address_id
    INNER JOIN city C ON B.city_id = C.city_id
    INNER JOIN country D ON C.country_ID = D.country_ID
  )
  GROUP BY country
  ORDER BY COUNT(customer_id) DESC
  LIMIT 10
)
GROUP BY city, country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY B.customer_id, B.first_name, B.last_name, D.city, E.country
ORDER BY average_amount_paid DESC
LIMIT 5
```

1.2) Inner statement/subquery & data output of the top 5 customers from the top 10 cities who've paid the highest total amounts to Rockbuster

```
SELECT
  B.customer_id,
  B.first_name AS customer_first_name,
  B.last_name AS customer_last_name,
  D.city,
  E.country,
  SUM(A.amount) AS total_amount_paid,
  COUNT(A.payment_id) AS number_of_payments
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE D.city IN (
  SELECT city
```

```

FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
WHERE D.country IN (
SELECT country
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
GROUP BY country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY city, country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY B.customer_id, B.first_name, B.last_name, D.city, E.country
ORDER BY total_amount_paid DESC
LIMIT 5;

```

Data Output:

C_ID	First_n	Last_n	City	Country	Total	NoP
537	"Clinton"	"Buford"	"Aurora"	"United States"	39.92	8
587	"Sergio"	"Stanfield"	"Celaya"	"Mexico"	34.93	7
367	"Adam"	"Gooch"	"Adoni"	"India"	34.93	7
84	"Sara"	"Perry"	"Atlixco"	"Mexico"	34.93	7
443	"Francisco"	"Skidmore"	"So Leopoldo"	"Brazil"	29.94	6

1.3) Main query & subquery put together incl. alias “average”

```

SELECT
AVG(total_amount_paid) AS average
From
(SELECT
B.customer_id,
B.first_name AS customer_first_name,
B.last_name AS customer_last_name,
D.city,
E.country,
SUM(A.amount) AS total_amount_paid,
COUNT(A.payment_id) AS number_of_payments
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE D.city IN (
SELECT city
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
WHERE D.country IN (
SELECT country
FROM customer A

```

```

INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
GROUP BY country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY city, country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY B.customer_id, B.first_name, B.last_name, D.city, E.country
ORDER BY total_amount_paid DESC
LIMIT 5
) AS total_amount_paid

```

Data Output:

Average
107.3540000000000000

1.4) CTE conversion

```

WITH average_cte (average) AS
(SELECT
  AVG(total_amount_paid) AS average
From
(SELECT
  B.customer_id,
  D.city,
  E.country,
  SUM(A.amount) AS total_amount_paid,
  COUNT(A.payment_id) AS number_of_payments
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE D.city IN (
  SELECT city
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
WHERE D.country IN (
  SELECT country
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
GROUP BY country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY city, country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY B.customer_id, D.city, E.country
ORDER BY total_amount_paid DESC
LIMIT 5

```

```

) AS total_amount_paid)
SELECT
average
FROM average_cte
GROUP BY average
ORDER BY average DESC

```

2) *How many of the top 5 customers are based within each country*

2.1) *Outer statement that counts the number of customers living in each country*

```

SELECT E.country, COUNT(DISTINCT B.customer_id) AS all_customer_count
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
GROUP BY E.country
ORDER BY all_customer_count DESC
LIMIT 10

```

2.2) *Inner statement that identifies the countries of the top 5 customers !*

```

SELECT E.country AS country, COUNT(DISTINCT B.customer_id) AS all_customer_count, COUNT(DISTINCT
top_5_customers.customer_id) AS top_customer_count
FROM customer B
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
LEFT JOIN
(SELECT
B.customer_id,
B.first_name AS customer_first_name,
B.last_name AS customer_last_name,
D.city,
E.country,
E.country_id,
SUM(A.amount) AS total_amount_paid,
COUNT(A.payment_id) AS number_of_payments
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE D.city IN (
SELECT city
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
WHERE D.country IN (
SELECT country
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
GROUP BY country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
)
GROUP BY city, country

```

```

ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY B.customer_id, B.first_name, B.last_name, D.city, E.country, E.country_id
ORDER BY total_amount_paid DESC
LIMIT 5
) AS top_5_customers ON top_5_customers.country_id = E.country_id
GROUP BY E.country
ORDER BY all_customer_count DESC
LIMIT 10;

```

Data Output:

COUNTRY	ALL_CUSTOMER_COUNT	TOP_CUSTOMER_COUNT
"India"	60	1
"China"	53	0
"United States"	36	1
"Japan"	31	0
"Mexico"	30	2
"Brazil"	28	0
"Russian Federation"	28	0
"Philippines"	20	0
"Turkey"	15	1
"Indonesia"	14	0

2.3) CTE conversion

```

WITH top_country_cte (country,
all_customer_count,
top_customer_count) AS
(SELECT E.country AS country, COUNT(DISTINCT B.customer_id) AS all_customer_count,
COUNT(DISTINCT top_5_customers.customer_id) AS top_customer_count
FROM customer B
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
LEFT JOIN
(SELECT
B.customer_id,
B.first_name AS customer_first_name,
B.last_name AS customer_last_name,
D.city,
E.country,
E.country_id,
SUM(A.amount) AS total_amount_paid,
COUNT(A.payment_id) AS number_of_payments
FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE D.city IN (
SELECT city
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
WHERE D.country IN (
SELECT country

```

```

FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_ID = D.country_ID
GROUP BY country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY city, country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
)
GROUP BY B.customer_id, B.first_name, B.last_name, D.city, E.country, E.country_id
ORDER BY total_amount_paid DESC
LIMIT 5
) AS top_5_customers ON top_5_customers.country_id = E.country_id
GROUP BY E.country
ORDER BY all_customer_count DESC
LIMIT 10)
SELECT
country,
all_customer_count,
top_customer_count
FROM top_country_cte
GROUP BY country, all_customer_count, top_customer_count
ORDER BY all_customer_count DESC

```